09 Variance Decomposition - Attribution of Performance Variation

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0.1 Introduction

Central output of V2: a variance decomposition attributing performance variation to each analytical decision. Published sepsis studies vary across multiple factors simultaneously (Cohen et al., 2024); this analysis varies one factor at a time.

Five factors:

Factor	Levels
Model class	Discrete-time hazard, Cox PH, GBT, NEWS2
Label operationalisation	Variant A (Narrow), B (Seymour-Standard), C (Liberal)
Validation split	Temporal, random
Prediction-time anchoring	Treatment-anchored, unrestricted
Evaluation metric	Utility Score, calibration, AUROC, AUPRC

Six pre-registered hypotheses structure the interpretation:

#	Claim	Threshold
H1	Label-variant spread $\geq$ model-class spread in AUROC	label spread $\geq$ model spread
H2	Coefficient instability across labels	≥ 1 covariate changes sign

#	Claim	Threshold
H3	Treatment-anchored AUROC < unrestricted	$\Delta \geq 0.03$
H4	Temporal-split AUROC < random-split	$\Delta \geq 0.02$
H5	Utility degradation > AUROC degradation (internal → external)	ratio comparison
H6	$\lvert\text{primary} - \text{GBT}\rvert \leq 0.02$ AUROC	$\lvert\Delta\rvert \leq 0.02$

0.2 Load Metric Results

Metric results from 07, external validation from 08, and the coefficient table from 05 are loaded.

```
## Metric results loaded: 36 rows
```

0.3 Variance by Factor

Each factor's spread is the range of AUROC (or Utility Score) across that factor's levels, with others held at reference values (Variant B, primary model, temporal split, unrestricted).

0.4 The Variance Decomposition Table

The study's primary deliverable. Each row holds four of the five analytical factors at a reference level and varies the fifth, so the number in `spread_auroc` is what that one decision is worth on its own. The reference levels are named in the factor column rather than left implicit, because a spread only means something once the reader knows what was held still.

Read the rows against each other, not in isolation. The point of the table is the ordering: whether the choice of model moves reported performance further than the choice of label, the split or the metric. The `hypothesis` and `h_threshold` columns record which pre-registered claim each row bears on.

Spreads here are point estimates with nothing attached. The intervals, and the multiplicity-corrected verdicts that supersede the descriptive ones below, come from stage 12.

0.5 Hypothesis Test Verdicts

Each hypothesis is evaluated against its threshold. "CONFIRMED" = effect meets/exceeds threshold; "NOT confirmed" = falls short. These are descriptive verdicts, not formal statistical tests.

```
## H1 (Label dominance): label_spread=0.012, model_spread=0.150 <U+2192> NOT confirmed

## H2 (Coefficient instability): 2 sign-changes: bilirubin_total, vaso_anyTRUE <U+2192> CONFIRME

## H4 (Split optimism): random <U+2212> temporal AUROC <U+0394>=-0.017 <U+2192> NOT confirmed (<

## H6 (Model-class null): |primary <U+2212> GBT| AUROC=0.017 <U+2192> NOT rejected (interpretabl

##
## NOTE: 09_hypothesis_verdicts holds point-estimate pre-checks only.
##       The confirmatory verdicts are 12_confirmatory_tests.
```

Table 1: All evaluation results (from 07_metrics_suite.Rmd)

variant	model	eval_window	auroc	auprc	utility_normalised	utility_max	alert_burden	l
A	primary	unrestricted	0.7713	0.0093	0.0000	0.0256	Inf	
A	news2	unrestricted	0.6413	0.0032	-0.5093	-0.0006	236.36	
A	cox	unrestricted	0.6589	0.0035	-4.7900	-0.0001	356.42	
A	gbt	unrestricted	0.7455	0.0088	-1.6141	0.0376	192.07	
A	qsofa	unrestricted	0.6012	0.0027	-2.2935	-0.2627	343.10	
A	sirs	unrestricted	0.6157	0.0032	-3.2088	-0.0349	346.95	
A	primary	treatment_anchored	NA	NA	0.0000	0.0001	Inf	
A	news2	treatment_anchored	NA	NA	-0.0817	-0.0003	Inf	
A	cox	treatment_anchored	NA	NA	-1.3545	0.0006	Inf	
A	gbt	treatment_anchored	NA	NA	-0.5427	0.0181	Inf	
A	qsofa	treatment_anchored	NA	NA	-0.6299	-0.0393	Inf	
A	sirs	treatment_anchored	NA	NA	-0.9267	-0.0020	Inf	
B	primary	unrestricted	0.7590	0.0096	0.0000	0.0252	Inf	
B	news2	unrestricted	0.6415	0.0032	-0.4911	-0.0004	246.72	
B	cox	unrestricted	0.6444	0.0033	-4.7204	-0.0001	361.57	
B	gbt	unrestricted	0.7424	0.0087	-1.7810	0.0313	196.53	
B	qsofa	unrestricted	0.6091	0.0028	-2.1935	-0.2522	316.82	
B	sirs	unrestricted	0.6114	0.0030	-3.0857	-0.0345	351.67	
B	primary	treatment_anchored	NA	NA	0.0000	0.0058	Inf	
B	news2	treatment_anchored	NA	NA	-0.0629	-0.0003	Inf	
B	cox	treatment_anchored	NA	NA	-1.2985	-0.0001	Inf	
B	gbt	treatment_anchored	NA	NA	-0.5353	0.0113	Inf	
B	qsofa	treatment_anchored	NA	NA	-0.6042	-0.0339	Inf	
B	sirs	treatment_anchored	NA	NA	-0.8863	-0.0023	Inf	
C	primary	unrestricted	0.7695	0.0201	0.0000	0.0954	Inf	
C	news2	unrestricted	0.6483	0.0074	-0.0985	0.0012	102.19	
C	cox	unrestricted	0.6733	0.0085	-1.6133	0.0025	148.66	
C	gbt	unrestricted	0.7550	0.0188	-0.5571	0.1110	89.19	
C	qsofa	unrestricted	0.6012	0.0060	-0.7045	-0.0476	150.86	
C	sirs	unrestricted	0.6242	0.0070	-1.0653	-0.0002	148.36	
C	primary	treatment_anchored	NA	NA	0.0000	0.0069	Inf	
C	news2	treatment_anchored	NA	NA	0.0016	0.0077	Inf	
C	cox	treatment_anchored	NA	NA	-0.5417	0.0082	Inf	
C	gbt	treatment_anchored	NA	NA	-0.1702	0.0361	Inf	
C	qsofa	treatment_anchored	NA	NA	-0.2474	-0.0020	Inf	
C	sirs	treatment_anchored	NA	NA	-0.3737	0.0035	Inf	

Table 2: THE VARIANCE DECOMPOSITION TABLE

factor	spread_auroc	spread_u
Model class (label=B, temporal, unrestricted)	0.1499	0.
Label operationalisation (model=primary, temporal, unrestricted)	0.0123	0.
Prediction-time anchoring (model=primary, variant=B, temporal)	NA	
Validation split (temporal vs random, model=primary, variant=B)	-0.0166	
Metric (AUROC vs Utility, internal<U+2192>external, model=primary, variant=B)	0.0630	
Model-class null |primary <U+2212> GBT| (variant=B, temporal, unrestricted)	0.0166	

0.6 Variance Decomposition Plots

Label-variant effect vs. model-class effect on AUROC. If H1 is confirmed, the label-variant spread exceeds the model-class spread.

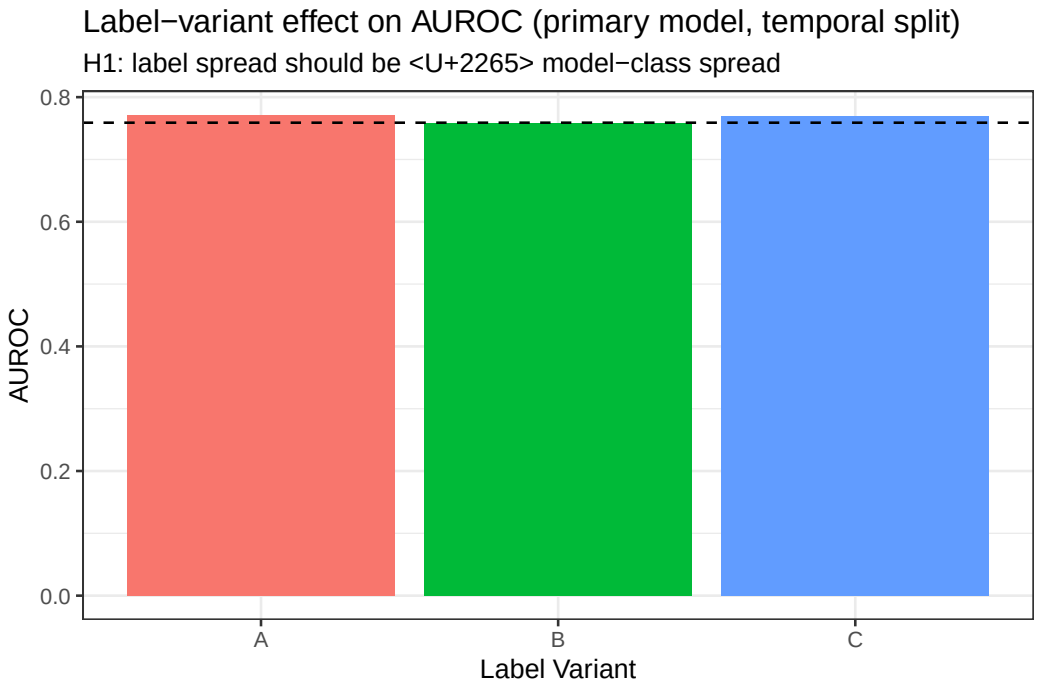


Figure 1: Variance decomposition: label-variant and model-class effects on AUROC.

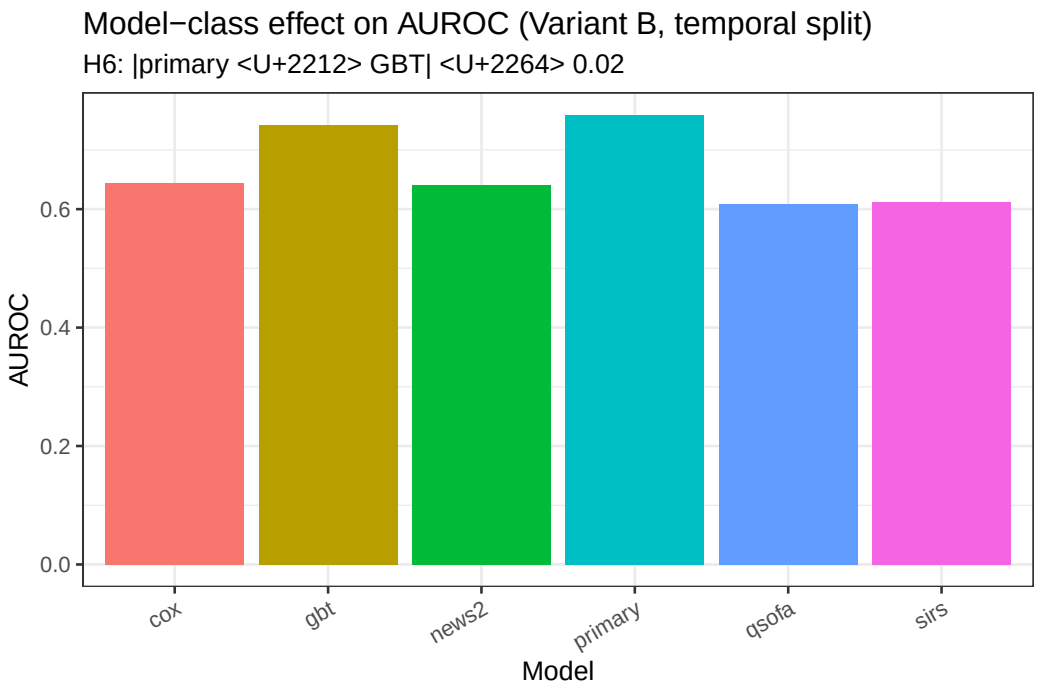


Figure 2: Variance decomposition: label-variant and model-class effects on AUROC.

```
##
## Variance decomposition complete.
```

0.7 Label-Anchor Attribution (Protocol Amendment 2)

Everything above varies the label *within* Sepsis-3. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic trigger, so the label spread entering H1 is partly a **definitional** quantity: it measures how much the reading of a definition matters, not how much the choice of construct matters. Whether the physiology itself carries the signal cannot be settled inside that family.

Sepsis-3 is a conjunction — *suspected infection* **and** *acute organ dysfunction*. This section splits it and fits the same model to each limb alone:

Variant	Suspicion anchor	Organ dysfunction	Reads a treatment decision?
B (Seymour-Standard)	required	required	yes
E (Anchor-Only)	required	not required	yes
D (Deterioration)	not required	required	**no**

Variant D is the treatment-independent labelling the decomposition was missing: its onsets are derived from recorded physiology alone, so no antibiotic order, culture draw or drug administration enters the label.

Two quantities are read off the comparison. The **anchor-attributable fraction** is the share of Variant B's above-chance discrimination that is reproduced by predicting the treatment anchor on its own,

$$\phi_{\text{anchor}} = \frac{\text{AUROC}_E - 0.5}{\text{AUROC}_B - 0.5},$$

and the **label agreement** statistics report how far the alternative labels move the onsets themselves.

```
##
## Anchor-attributable fraction (primary model): 104.1% of Variant B's above-chance AUROC is rep
```

0.8 Label Agreement Between Definitions

How far the alternative labels move the onsets, measured on the same stays: the share of stays each pair labels identically (Cohen's κ for chance correction) and, among stays positive under both, the median shift in onset hour. A label that disagrees with Sepsis-3 about *which* patients deteriorate is a different construct, not a re-reading of the same one.

0.9 The Pre-Treatment Window (H3), Analytically and Empirically

H3 asks what happens when evaluation is restricted to the hours before the first antibiotic or culture. Under variants A–C the answer is fixed before any data are seen, and it is worth stating the argument as a proposition rather than as a result.

> **Proposition.** Let t_{anchor} be the first qualifying antibiotic order or culture draw of the admission and let t_{onset} be the onset assigned by variants A, B, C or E. Then $t_{\text{onset}} \geq t_{\text{anchor}}$ for every labelled stay, so the pre-anchor window contains no onsets.
>
> *Proof.* Under E, $t_{\text{onset}} = t_{\text{susp}}$, and t_{susp} is by definition the later of a qualifying antibiotic-culture pair, hence $t_{\text{susp}} \geq t_{\text{anchor}}$. Under A, B and C the SOFA criterion gates *membership* only; the assigned onset is still t_{susp}. The inequality therefore carries over unchanged. □
>
> The observed zero is a consequence of this implementation of the label, not a statement about MIMIC-IV.

The empirical question is what a definition *without* that property records in the same window. Two variants supply it. Variant D drops the infection anchor entirely. Variant F (Protocol Amendment 4) keeps the Sepsis-3 conjunction exactly — identical membership to Variant B — and restores only the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ timing of Seymour et al. (2016) and the PhysioNet/CinC 2019 Challenge, under which an onset may precede the anchor whenever the organ-dysfunction rise is detected first.

```
##
## Pre-treatment onsets: B (conjunction timing): 0 in 179,003 person-h; F (standard min(t_susp,
## Variant F moved 1,738 of 6,447 onsets earlier (27.0%), median shift -14 h.
```

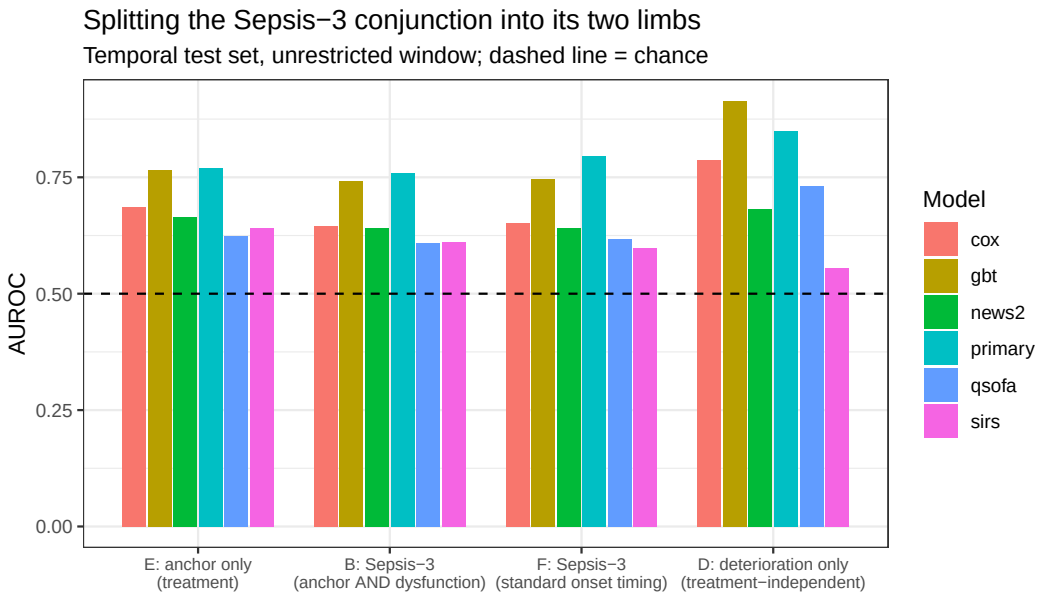


Figure 3: Discrimination under each limb of the Sepsis-3 conjunction. Variant B requires suspected infection and organ dysfunction; E keeps only the treatment anchor; D keeps only the physiological deterioration and contains no treatment timestamp; F keeps the full conjunction but times the onset as min(t_susp, t_SOFA).

0.10 Reading the Attribution

Three things follow, and only the third is empirical rather than definitional.

1. **Variant E is not a sepsis model.** Its label is the moment a clinician sent a culture and started antibiotics. Whatever discrimination it achieves is discrimination about treatment behaviour, and ϕ_{anchor} states how much of Variant B's apparent skill that behaviour already accounts for.
2. **Variant D is not directly comparable to B on AUROC.** Its label is a threshold function of recorded physiology, and five of the six SOFA components (all but PaO_2/FiO_2) are model inputs, so a high AUROC would be partly self-fulfilling. The comparison that carries information is not the level but whether the label changes *which* patients are flagged, which the agreement table answers.
3. **The label-sensitivity finding survives only to the extent that D and B disagree.** If a treatment-independent label picks out substantially different patients at substantially different hours, then label choice is a real constraint on what a sepsis model can be trained to do, not an artefact of how one definition is read.

```
## Primary model AUROC: B (Sepsis-3): 0.759 | E (anchor only): 0.770 | D (deterioration): 0.849
```

```
## Variant D vs B: kappa = 0.057, 55.5% of B's septic stays also deteriorate, median onset shift
```